

2024 September Floods and Landslides

Situation Report #2 as of 1 October 2024

(First SitRep published on 29 September 2024)



Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and
Management Authority

This report is produced by National Disaster Risk Reduction and Management Authority (NDRRMA) in collaboration with sectoral ministries, departments, provincial and local governments. It covers the situation of floods and landslides due to the heavy rainfall across the country from 26-28 September.

Summary of Loss and Damage

Human Casualties



224
Deaths



158
Injury



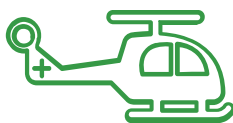
24
Missing

Emergency Rescue



30,731

Security Personnel Deployed



713 Heli Rescued
Stranded Passengers



12,586
Rescued

Private Housing and Critical Infrastructure



700+

Fully Damaged



500+

Partially Damaged



43

Schools



41

Health Center

Infrastructure



11 Hydropower
625.96 MW
NPR 3 Billion Loss



6 Irrigation Projects
NPR 1.35 Billion
Loss



27 Highway Blocked
NPR 1.5 Billion
Loss



14 Damaged
Suspension Bridge



11 Damaged
Bailey, RCC, ARC Bridge
1.02 Billion Loss

Telecommunications



302 Sites Down



NPR 165 Million Loss
Nepal Telecom



NPR 70 Million Loss
NCELL



Internet Service Providers
NPR 0.31 Billion Loss

Agriculture and Livelihood



Paddy Field
88,476 Hectares
NPR 3.5 Billion Loss



Fruits & Vegetables Farm
4,765 Hectares



Sugarcane Field
85 Hectares
NPR 0.34 Billion Loss



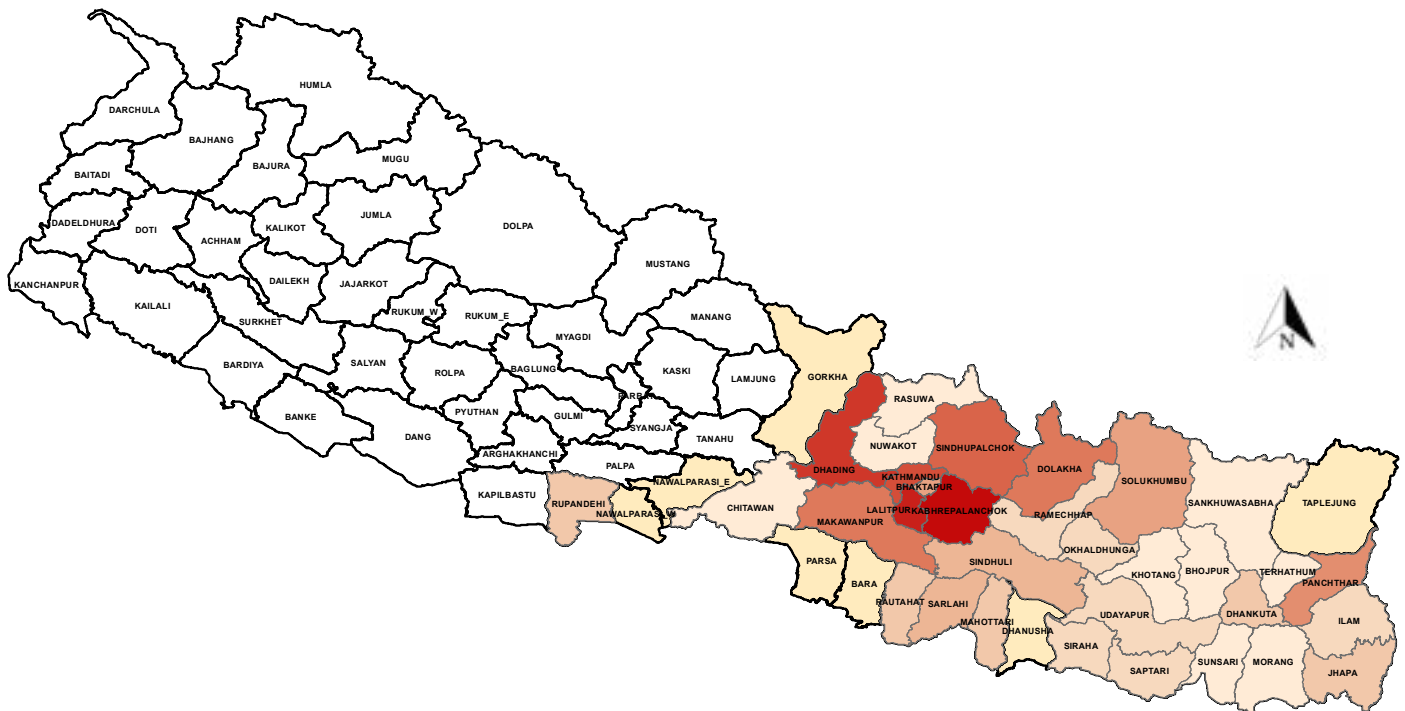
Fish Pond
458 Hectares
NPR 1.13 Billion Loss



Livestock
26,205



Cardamom Field
1,300 Hectares
NPR 0.7 Billion Loss



Map 1: Map of Nepal showing affected districts

Highlights

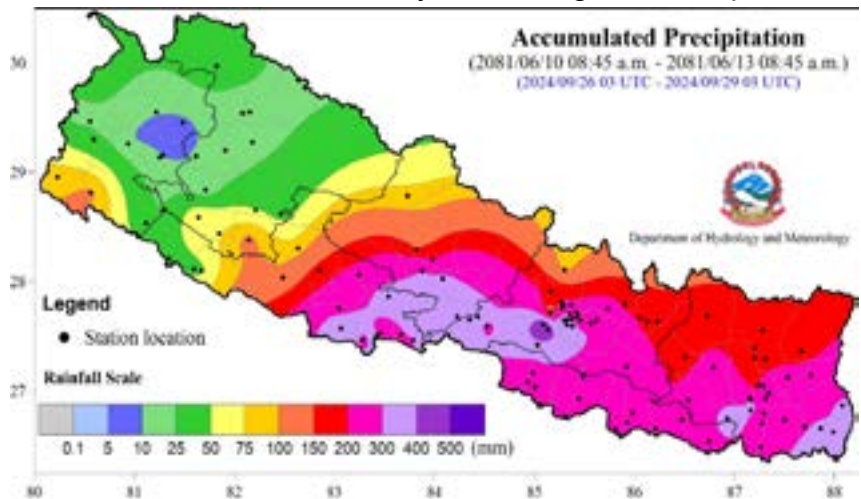
- 1) As of 1 October, the total number of **deaths has reached 224**, with **24 individuals still missing** and **158 reported injured**.
- 2) As of 30 September, Army helicopters have **rescued 713 stranded passengers**, and more than **12,586 people have been rescued** through various rescue operations that deployed 30,731 security personnel until 1 Oct.
- 3) The Ministry of Energy, Water Resources, and Irrigation estimates the total damage to be approximately **NPR 4.35 billion** that includes **NRS 1.35 billion river training works** and irrigation schemes, and around **NPR 3 billion damage in hydropower projects**. 11 operational hydropower projects with a total installed capacity of 625.96 MW have been damaged. A total of five transmission lines, along with several high-tension and low-tension lines and transformers, have been damaged, including the most recent high-tension damage in Nawalparasi (Bardaghat Susta Paschim).
- 4) According to a preliminary report from the Department of **Roads**, the combined damage to roads and bridges has resulted in a total **loss of NPR 2.5 billion**. Of this, NPR 1.50 billion is attributed to road damage, while the damage to bridges amounts to NPR 1.02 billion. The floods and landslides have also **damaged 11 bridges**, including three Bailey bridges, and transportation has been halted at 33 locations across 11 major highways.
- 5) According to the report from the Ministry of Education, **43 school buildings** have been damaged **across 10 districts**.
- 6) According to the Ministry of **Health**, a total of **41 health facilities** have sustained damage due to floods and landslides across the country.
- 7) The agriculture and livestock sector has reported **losses exceeding NPR 6 billion**, according to preliminary reports from the Ministry of Agriculture and Livestock Development.
- 8) The Ministry of Communication and Information Technology has estimated the loss in the telecommunications sector to be **NPR 0.31 billion**, which includes losses incurred to private internet service providers.
- 9) The critical **highways connecting Kathmandu** are now operational.

- 10) The Executive Committee meeting of 30 September decided to expedite the construction of temporary shelters by **providing NPR 50,000 to each affected household**, in line with the Guidelines for the provision of temporary housing construction grants to families affected by disasters, 2081 B.S. The committee also requested support from humanitarian organizations in Nepal for additional assistance. Later that day, the Cabinet approved a **relief package of NPR 200,000** each for the families of deceased persons .

Situation Overview

Continuous rainfall between 26-28 September, 2024 resulted in multiple incidents of flood and inundations, small, medium and large scale landslides across the country. According to the Department of Hydrology and Meteorology (DHM), out of the 222 measurement stations where rainfall was simultaneously recorded across the country, 77 stations reported heavy rainfall, with amounts exceeding 200 mm on 28 September.

Among the hydrological gauging stations installed in various rivers across the country, **23 stations** recorded water levels surpassing the danger level, and an additional 14 stations recorded water levels exceeding the warning level. The water flow in the Saptakoshi River has reached the highest level in 56 years.



After 22 years, Kathmandu witnessed record rainfall on Friday and Saturday. DHM reported that the rainfall on Saturday was the highest ever in the valley's recorded history, surpassing the previous record set in 2002. Rainfall at the Tribhuvan International Airport station in Kathmandu reached **239.7 millimeters in 24 hours**, compared to the previous record of 177 millimeters in 2002.

Continuous heavy rainfall has caused widespread flooding and landslides across Nepal, resulting in **219 fatalities, 27 missing persons, and 143 injured**. Rescue operations have **saved 12,586 individuals** so far, and efforts to provide shelter, food, and essential supplies are ongoing. Security forces and local bodies are coordinating these relief efforts, with priority given to clearing landslide-blocked roads to facilitate the flow of relief materials. Major highways, including **Narayanghat-Muglin and the East-West Highway**, have reopened, though some sections remain under repair due to landslide damage.

Key Actions Taken

Key Decisions of Cabinet Meeting

- A three-day mourning period declared, the national flag to be flown at half-mast.
- Submit a detailed report by 1 October on the loss of lives and properties, along with cost estimates for reconstruction.
- The government to provide Rs 2,00,000 to the families of those who died in the floods and landslides.
- The family of a missing person to be provided with the same compensation as the family of a deceased person if the missing person is not found within ten days.
- Free treatment to be provided for all individuals injured in the disaster.
- NRS 1 billion to be transferred immediately to Disaster Management Fund.

Key Decisions of 25th Executive Committee Meeting held at NDRRMA on 30 September

- Continue search and rescue operations in disaster-affected areas.
- Initiate relief fund distribution through a one-door system.
- Free treatment for injured individuals across affected districts and municipalities.
- Assign LDMCs to identify beneficiaries for temporary housing construction, with Rs 50,000 provided in two installments as per the government standard.
- Clear blocked roads and assist stranded passengers.
- Conduct public health programs and awareness campaigns in areas at risk of epidemics.
- Mobilize humanitarian aid from UN agencies, Nepal Red Cross, and partner organizations as needed.
- Assess damages from floods, landslides, and heavy rains.
- Direct sectoral ministries to begin reconstruction of damaged public infrastructure.
- Urge provincial and local disaster management committees to carry out post-disaster operations as per legal responsibilities.

Key Decisions of Bagmati Province Executive Committee Meeting

- The Bagmati Province Executive Committee meeting on 30 September decided to mobilize all sectoral ministries to collect data on losses and damages in various sectors, including housing, health, and agriculture, and to formulate plans based on this data.
- The committee also allocated NPR 23 million to the Province Disaster Management Fund for distribution to District Administration Offices in several districts, including NPR 6 million for Kavre, NPR 5 million for Dhading, NPR 3 million for Lalitpur, and NPR 2 million each for Kathmandu and Sindhuli. An additional NPR 1 million was allocated to Dolakha, Bhaktapur, Ramechhap, and Makawanpur.

Housing and Settlement

Province	Fully Affected HHs	Partially Affected HHs	No. of displaced HHs	Affected HHs
Koshi Province	195	200	324	1777
Madesh Province	200	36	96	488
Bagmati Province	123	164	488	1062

Damage to Highways and Roads

50 national highways across the country have been blocked, with transportation halted at 47 locations on 17 national highways as of 7:00 AM today. The major highways connecting the capital, such as Narayanghad-Muglin, Naubise-Muglin, and other roads affected by rivers and streams, have now been cleared, and all major highways are operational for traffic. Some sections of the East-West Highway that were blocked have been reopened for traffic.

- The **Nuwakot-Betrawati-Chhahare-Trishuli** segment of the Pasang Lhamu Highway, which was previously obstructed, is also now operational.
- **Ramnagar-Brahmadev-Dolalghat-Dhulikhel** section, which was affected by landslides, has been partially cleared and traffic has resumed.
- **Dhalkebar-Birgunj section of the East-West Highway is now open.** However, due to the damage caused by the flood to the **Mulkot Bridge** along the B.P. Highway, temporary arrangements have been made for vehicles to use a nearby alternative route.

- Efforts are also underway to clear other blocked highways, including Beni-Jomsom-Khobang, and Galchhi-Trishuli roads, and they are expected to be operational soon.

The Ministry of Physical Infrastructure and Transport is actively working to restore damaged roads and bridges and facilitate the movement of vehicles.

Damage to Drinking Water and Sanitation

Damages to Kathmandu Valley Drinking Water Limited (KUKL):

- Due to the flow of the Bagmati River, the water supply from the Bagmati River has been halted as both the old and new Sundarijal Water Treatment centers.
- The flood in the Manohara River severely impacted the Dugwell area, which has affected the water supply to Bhaktapur and Baneshwar areas.
- To prevent further damage to the drinking water distribution system, distribution activities have been suspended.
- Due to damage to the supply system of the Pharping Hydropower Water Reservoir, water supply has been disrupted in the Lalitpur area.
- Preliminary assessments indicate that property worth approximately 1 Billion NPR has been affected.

Damage to the Banepa Drinking Water

- Some damage has been reported at the Dhuneshwar Shashipani stream sources, and it may take 1-2 days to resume distribution.

Damage to the Panauti Drinking Water

- A preliminary report indicates damage to about 2 km of 90 mm pipes from Mulkhola to the Panauti tank.
- The pipeline from Haridol to the water tank has been completely damaged.

Damages in other locations

- **Ilam Urban Drinking Water and Sanitation Project:** Intake, stream crossing, and pipeline were damaged.
- **Bhojpur Urban Drinking Water and Sanitation Project:** Intake damaged.
- **Charikot Sewage Project (under construction):** Damage to the stone masonry wall *and stone soling road*.
- **Panchkhal Urban Drinking Water and Sanitation Project (under construction):** Intake, reservoir, stone masonry wall, main transmission and distribution pipeline, and electric poles were damaged.
- **Sunkoshi Lift Central Drinking Water Project (Ghumi, Katari):** Sump well, reservoir, sedimentation tank, and guard house were affected by the flood.
- **Dapcha Kashikhanda Drinking Water Project in Kavre District:** The source and filter unit at the Timale Drinking Water Project were reported to be covered by the flood.
- **Nanggedanda Drinking Water Project in Sindhuli:** Damage occurred to the sump well.
- The filter and 200 m³ water tank at Kashikhanda Drinking Water Project were washed away by the Roshi River.

Damage on Irrigation

Details	Estimated Loss
10 branch canals under the Sunsari-Morang Irrigation Project	NPR 50 Million
Bagmati Irrigation Project eastern and western main canals	NPR 54 Million
Jhajh Irrigation canal Project	NPR 50 Million
Koshi Pump-Chandra Canal Project	NPR 5 Million
Kamala Irrigation Management Project	NPR 50 Million
Narayani Irrigation Management Project	NPR 1 Million
Mahakali Irrigation Project in Kanchanpur	NPR 1.14 Billion
Total estimated loss and damage	NPR 1.35 Billion

The estimated cost of damage in irrigation is NPR 190 million, and the damage to the Sunkoshi Marin Diversion Multipurpose Project is yet to be assessed.

Status of Highways & Roads

S.N.	Province	District	Municipalities	Ward	Highway Name	Reason for obstruction	Status
1	Koshi	Panchthar	Phidim	7	Mechi Highway	Landslide	Closed
2	Koshi	Panchthar	Phidim	4,7	Mechi Highway	Bailey Bridge swept away	Closed
3	Koshi	Panchthar	Tumbewa	2	Tamar Corridor	High Water Flow	Closed
4	Bagmati	Makwanpur	Bhimphedi	8	Kantilokpath road section	Bailey bridge swept away	Closed
5	Bagmati	Makwanpur	Indra Sarovar	3	Bhimphedi-Sisneri Highway	Landslide	Closed
6	Bagmati	Makwanpur	Indra Sarovar	2	Kulekhani fakhel road section	Landslide	Closed
7	Bagmati	Makwanpur	Bhimphedi	4	Bhaise Bhimphedi	Landslide	Closed
8	Bagmati	Makwanpur	Indra Sarovar	1	Kulekhani-Markhu	Landslide	Closed
9	Bagmati	Ramechhap	Manthali	6	Selegat-Khurkot main road	Swept away by River	Closed
10	Bagmati	Kavre	Roshi	11	B.P Highway	Landslide	Closed
11	Bagmati	Kavre	Roshi	12	B.P Highway	Landslide	Closed
12	Bagmati	Kavre	Roshi	9	B.P Highway	Landslide	Closed
13	Bagmati	Kavre	Dhulikhel	8	Araniko Highway	Landslide	Closed
14	Bagmati	Sindhuli	Golanjar	6	Madhya Pahadi Lok-marga	Landslide	Closed
15	Bagmati	Sindhuli	Sunkoshi	6	B.P Highway	Landslide	Closed
16	Bagmati	Sindhuli	Sunkoshi	1,2	B.P Highway	Landslide	Closed
17	Bagmati	Sindhuli	Phikkal	2	Madhya Pahadi Lok-marga	Landslide	Closed
18	Bagmati	Dhading	Benighat Rorang	3	Prithivi Highway	Landslide	One way open
19	Bagmati	Dhading	Gajuri	2	Prithivi Highway	Dry landslide	One way open
20	Bagmati	Dhading	Dhunibesi	9	Prithivi Highway	Landslide	One way open
21	Bagmati	Dolakha	Tamakoshi	5	B.P Highway	Flood	Closed
22	Bagmati	Sindhupalchowk	Barabhise	5	Araniko Highway	Landslide	Closed

23	Bagmati	Sindhupal-chowk	Panch Pokhari Thangpal	8	Melamchi-Bhotang road section	Landslide	Closed
24	Bagmati	Sindhupal-chowk	Bhotekoshi	3	Barahbise-tatopani	Bailey bridge swept away	Closed
25	Bagmati	Sindhupal-chowk	Bhote Koshi	2	Araniko Highway	Flood	Closed
26	Bagmati	Sindhupal-chowk	Choutara Sangachowkgadi	9	Araniko Highway	Dry landslide	Closed
27	Lumbini	Nawalparasi West	Bardaghat	4	East-West Highway	Road sink	Closed

Immediate Needs and Priorities

- Rescue and relief for those stranded on highways due to flash floods and landslides.
 - Temporary shelter for displaced families.
 - Tents with light and other utilities for affected families.
 - Emergency water, sanitation and health facilities.
 - Provision of food and other humanitarian supplies.
 - Restoration of communication networks and electricity in affected areas.
 - Construction of temporary learning centers and health facilities.
 - Debris clearance and repair of disrupted road networks and highways.
- Immediate Needs and Priorities

Other Needs

- Prefabricated steel truss vehicular bridge – Total of 1050 meters
- Outboard motor with raft – 50 sets
- Heavy lifting drones (for relief material distribution) – 25 Unit
- Combi tools for road traffic accident (hydraulic) – 60 sets
- Flood light tower (Portable) – 100 units
- Power ascender with rope launcher – 30 sets

Technical Support: NHSRP



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